Specification: TW/THW CALTRANS
Issue Date: 12/31/09
Supersedes: 11/30/07


# Kris-Tech Wire Co.

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1.0 Scope: This specification describes single conductor polyvinylchloride (PVC) insulated, 600-volt dual rated type TW/THW. It is intended for use in accordance with the National Electrical Code (NEC) Article 310 and the California Dept. of Transportation (CALTRTANS) Specifications. It is suitable general purpose power wiring in residential, commercial and industrial applications with circuits at temperatures from -25°C to 75°C per NEC Article 310. This product complies with EU Directive 2002/95/EC known as ROHS.

2.0 Applicable Standards:
- 2.1 ASTM B-1, B-3, and B-8 for copper conductors.
- 2.2 Underwriters Laboratories (UL) Standards as follows: 83 (TW/THW) for Thermoplastic Insulated Wire.
- 2.3 CALTRANS traffic signal cables.

3.0 Conductors: The single copper conductors shall be solid or stranded annealed or hard uncoated copper per UL83 and ASTM requirements.

4.0 Insulation: The conductor shall be insulated with PVC as specified and applied tightly to the conductor in a concentric manner. The minimum at any point shall not be less than 90% of the specified average thickness in compliance with UL 83.

5.0 Identification: The wire shall be identified by surface or indent marking indicating the manufacturer, conductor size, voltage rating, UL symbol, and type designation.

6.0 Packaging: Standard packages are 500 ft/spool. – 4 per carton, 500 ft. knockout cartons, or 2500 feet/spool. Custom lengths are available.

7.0 Testing: The wire shall be continuously spark tested at 7500 Volts DC. Other electrical and mechanical tests shall be in accordance with UL 1581.

| AWG | Conductor Construction | Insulation (Mils) | Nominal Overall Diameter – Inches | Shipping Wgt. (Lbs./Mft.) |
|---|---|---|---|---|
| 14 | Solid or 7 strands | 45 | 0.16/0.17 | 23/24 |
| 12 | Solid or 7 strands | 45 | 0.17/0.19 | 32/33 |
| 10 | Solid or 7 strands | 45 | 0.20/0.21 | 46/47 |
| 8 | Solid or 7 strands | 60 | 0.25/0.27 | 73/75 |
| 6 | Solid or 7 strands | 60 | 0.28/0.31 | 105/107 |
| 4 | 7 or 133 strands | 60 | 0.35/0.39 | 160/167 |
| 2 | 7 or 133 strands | 60 | 0.41/0.45 | 250/265 |
| 1 | 19 strands | 80 | 0.54 | 340 |
| 1/0 | 19 strands | 80 | 0.62 | 410 |

**Non-standard STRANDING, COLORS, and CUSTOM PACKAGING are available by quotation.**